MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

SWITCH, ANALOG, KEY CODE FAG

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic switch module.

1.2 Classification. This specification covers one class of modules.

Class I as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787	-	Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389	-	Design Requirements for Standard Electronic Modules.
MIL-STD-1665	-	Test Equipment for the Standard Electronic Modules Program.

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20360, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:05Z
Check the source to verify that this is the current version before use.

3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 0.26 failures per million operating hours when predicted in accordance with MIL-STD-1389.

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be FAG.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/113-1, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance test. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

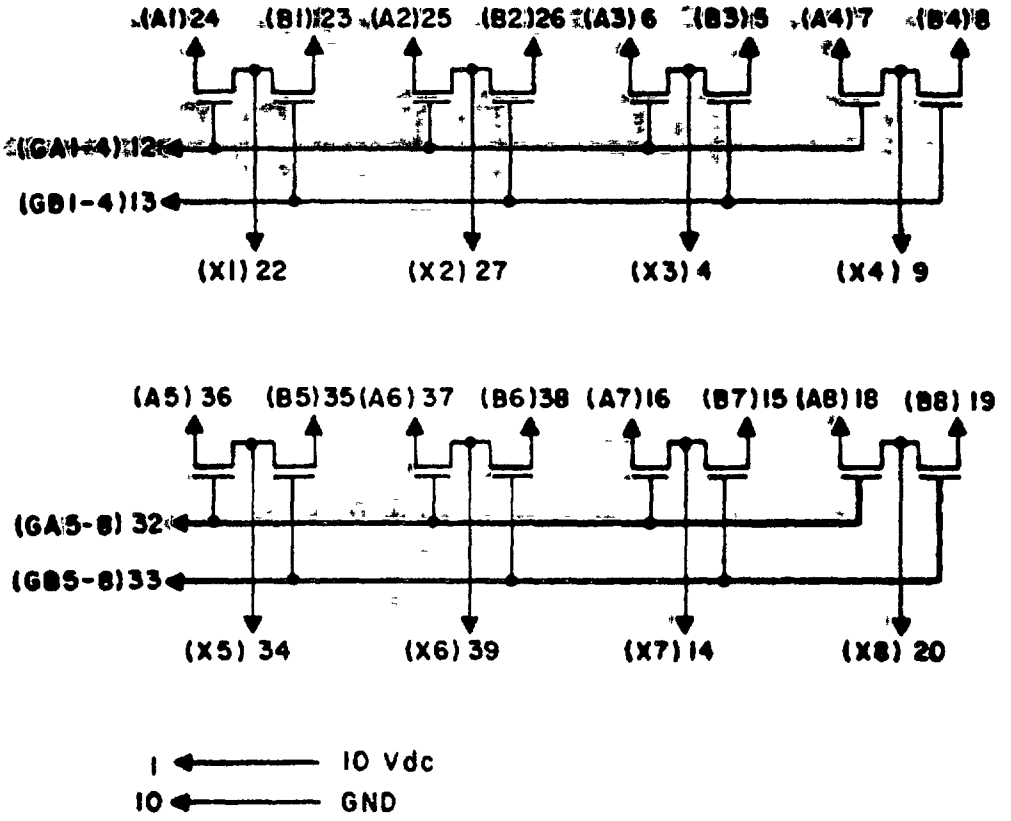
4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through P shall be performed in any order with the following exceptions. If test conditions A or B are part of a sequence of tests, the order of testing shall be as follows. A, B, where applicable, followed by the remaining tests in the sequence.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:05Z
Check the source to verify that this is the current version before use.



NOTE: A and B = Analog inputs
GA = Gate A input
GB = Gate B input
X = Analog output

FIGURE 1. FAG module function diagram, analog switch.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:05Z
Check the source to verify that this is the current version before use.

TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A1,A2	Isolation	II	S	10	∞	10	∞	10	∞	10	∞	MΩ
B1-B16	Stress	III	S	9.5	10.1	9.5	10.1	9.5	10.1	9.5	10.1	Vrms
C1-C16	ON resistance (600Ω) check	IV	X	-9.3	-10.1	-9.3	-10.1	-9.3	-10.1	-9.3	-10.1	V
C17-C32	ON resistance (200Ω) check	IV	X	9.7	10.1	9.7	10.1	9.7	10.1	9.7	10.1	V
D1-D16	OFF resistance check	V	X	0.0	1.0	0.0	1.0	0.0	1.0	0.0	1.0	µA
E1-E16	Analog input leakage current check	VI	X	0	50	0	50	0	100	0	100	mV
F1-F4	Gate input leakage current check	VII	X	0	10	0	10	0	20	0	20	mV
G1-G8	Output leakage current check	VIII	X	0	50	0	50	0	100	0	100	mV
H1-H16	Analog input capacitance	IX	Q	0	21	0	21	0	21	0	21	pF
J1-J4	Gate input capacitance	IX	Q	0	30	0	30	0	30	0	30	pF
K1-K8	Analog output capacitance	IX	Q	0	26	0	26	0	26	0	26	pF
L1-L8	Gate to output capacitance	IX	Q	0	16	0	16	0	16	0	16	pF
M	Power supply transient rise time	X	S	2.5	7.5	2.5	7.5	2.5	7.5	2.5	7.5	µs
N1-N8	Propagation delay time (t_{PLH})	XI	S	0	100	0	100	0	150	0	150	ns
P1-P8	Propagation delay time (t_{PHL})	XI	S	0	100	0	100	0	150	0	150	ns

1/ "X" parameters shall be 100 percent tested during module acceptance tests.
2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.
3/ "Q" parameters shall be tested during qualification testing only.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:05Z
Check the source to verify that this is the current version before use.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through P in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25° ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions C through G, and M through P shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 3.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column, and test condition M, shall be performed using the initial limits at 25° ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3. Following the life test and while still at the maximum class temperature, electrical tests C through P shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests C through P shall be performed in the order specified in 4.4.2 using the 25° ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

6.3 Cross reference of classification. Military Specification MIL-M-28787/113 supersedes Department of the Navy, Naval Ordnance Systems Command Drawing No. 10001-2833693 dated 20 September 1972. Part number M/28787/113-1 supersedes part numbers 2833693 and M28787/113. These parts are interchangeable.

6.4 Module function. This module contains two independent 4-pole double-throw analog switches. The switch performs the function of a 4-pole double-throw switch as shown on figure 2.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:05Z
Check the source to verify that this is the current version before use.

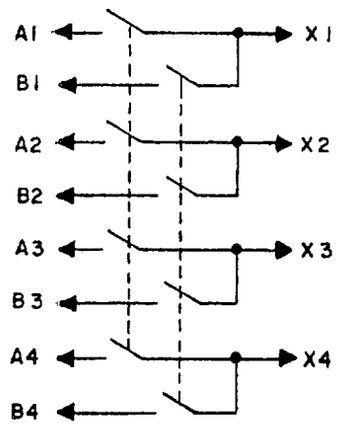


FIGURE 2. Equivalent circuit, 4-pole double-throw analog switch.

6.5 Module description. The following electrical parameters reflect the 0° to 60°C end-of-life parameters from table I.

Supply voltage	+10 Vdc, + 14 Vdc absolute maximum
Speed	Gate switching 1 MHz maximum
Maximum power dissipation	80 mW (200 mW per circuit)
Analog input voltage	+10 V peak maximum for a supply voltage of + 10 Vdc
Analog input leakage current	1 µA maximum
Gate voltage	+10 V to -30 V maximum for power supply = +10 Vdc
Gate threshold	-1.0 V GATE TO SOURCE
ON resistance	600 Ω maximum (V_{IN} = -10 V, Gate = -20 V)
	200 Ω maximum (V_{IN} = +10 V, Gate = -20 V)
OFF resistance	>10 MΩ
Output leakage current	1 µA maximum

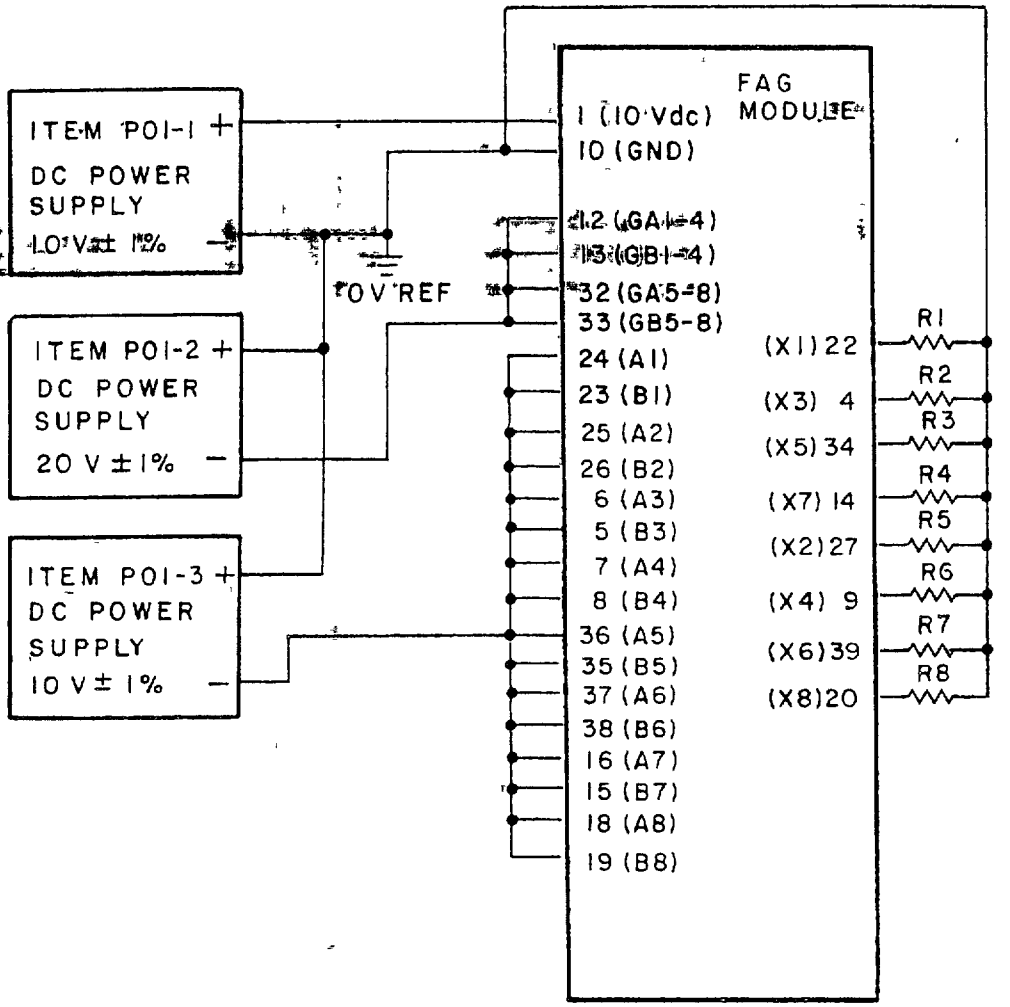
6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Agent:
DLA - ES

Preparing activity
Navy - EC

(Project 5963-N202)

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:05Z
Check the source to verify that this is the current version before use.



NOTE. R1-R8 = 1 kΩ ±1%, 1/8 W.

FIGURE 3 Operating temperature, component temperature, and life test circuit.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:05Z
Check the source to verify that this is the current version before use.

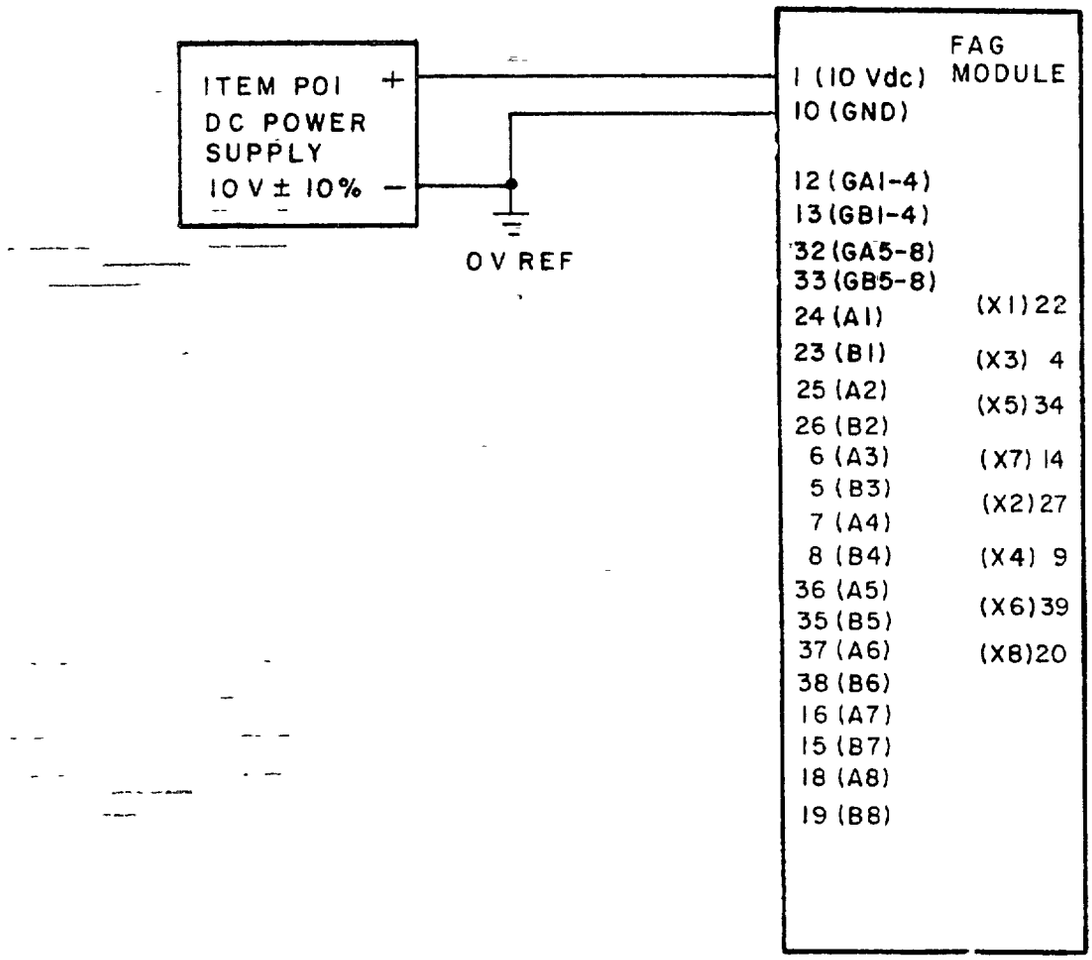


FIGURE 4. Vibration test circuit

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:05Z
Check the source to verify that this is the current version before use.

TABLE II. Isolation test.

Test condition	Parameter	Connect megohm bridge item R06		Initial limits 25°±5°C ambient		Unit
		+ terminal	- terminal	Min	Max	
A1	Isolation	All pins	Module frame 1/	10	∞	MΩ
A2	Isolation	All pins except 2,3,11,17,21,28, 29,30,31,40	2,3,11,17,21, 28,29,30,31,40	10	∞	MΩ

1/ Penetrate the anodized finish on the module frame with a sharply pointed probe. Connect the negative lead of the megohm bridge to the probe. The megohm bridge output shall be set at 100 ±10 Vdc.

WARNING

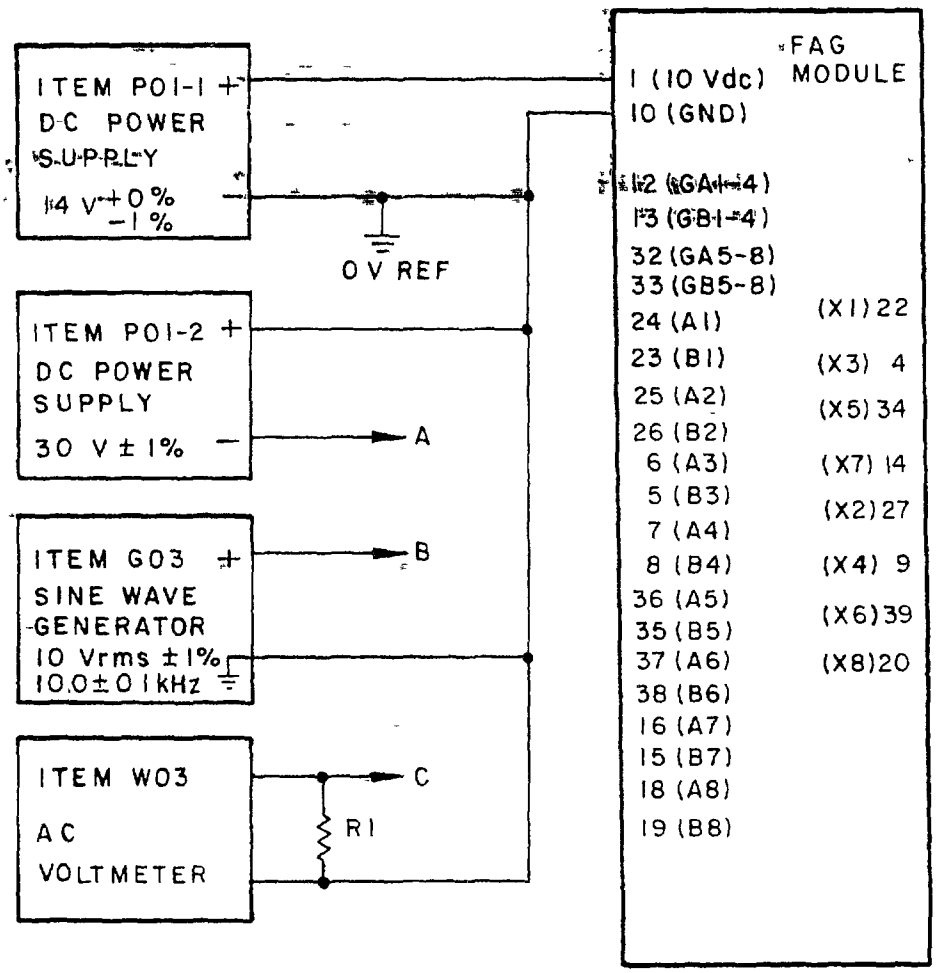
When using the megohm bridge, high voltage may be present at any of the output binding posts. Always set the function switch of the megohm bridge at "DISCHARGE" before connecting or disconnecting the test leads.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:05Z
Check the source to verify that this is the current version before use.

TABLE III. Stress test

Test condition	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
B1	Stress	12	24	22	9.5	10.1	Vrms	5
B2	↓	13	23	22	↓	↓	↓	↓
B3		13	26	27				
B4		12	25	27				
B5		12	6	4				
B6		13	5	4				
B7		13	8	9				
B8		12	7	9				
B9		32	36	34				
B10		33	35	34				
B11		33	38	39				
B12		32	37	39				
B13		32	16	14				
B14		33	15	14				
B15		33	19	20				
B16	↓	32	18	20	↓	↓	↓	↓

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:05Z
Check the source to verify that this is the current version before use.



NOTES:
1. R1 = 100 kΩ ±1%, 1/8 W.
2. If test condition is successively repeated, the time interval between execution of each of these test conditions should be a minimum of 10 seconds in order to preclude detrimental effects due to module heating.

FIGURE 5. Stress test circuit for table III.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:05Z
Check the source to verify that this is the current version before use.

TABLE IV. ON resistance check test.

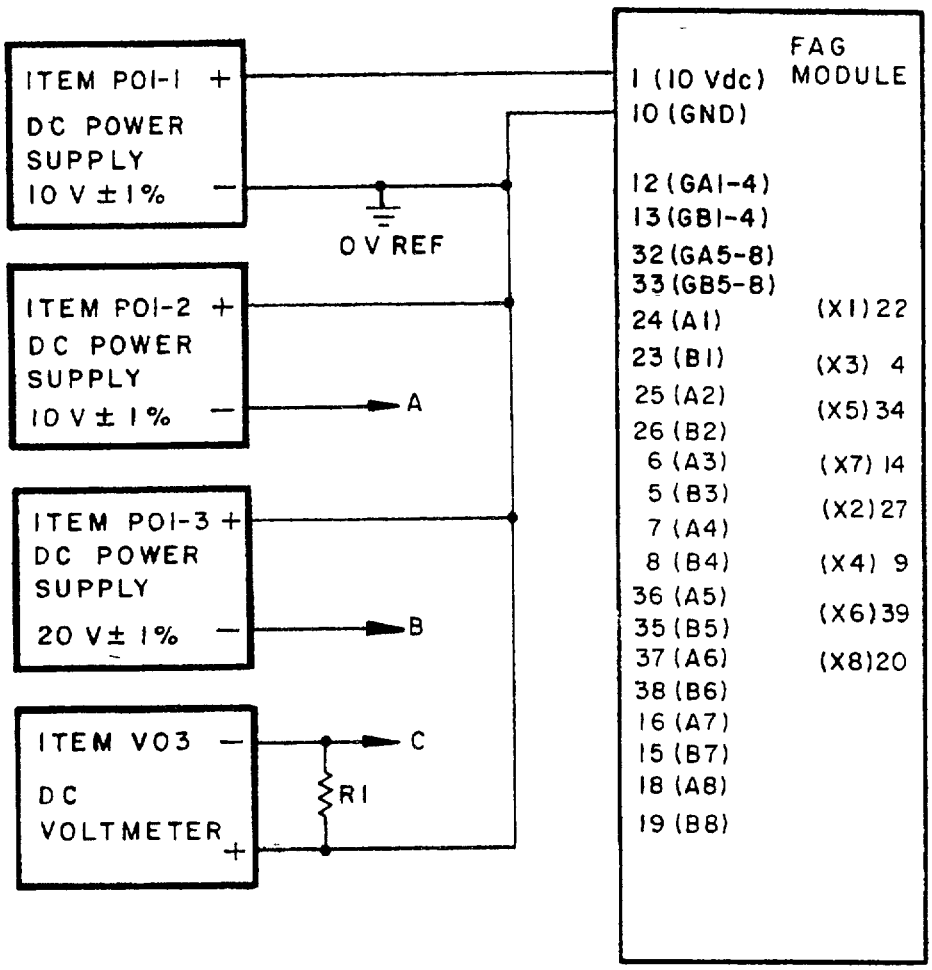
Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
C1	ON resistance (600Ω) check	24	12	22	-9.3	-10.1	V	6
C2		6		4				
C3		25		27				
C4		7		9				
C5		36	32	34				
C6		16		14				
C7		37		39				
C8		18		20				
C9		23	13	22				
C10		5		4				
C11		26		27				
C12		8		9				
C13		35	33	34				
C14		15		14				
C15		38		39				
C16		19		20				

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:05Z
Check the source to verify that this is the current version before use.

TABLE IV. ON resistance check test - Continued

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
C17	ON resistance (200Ω) check	24	12	22	9.7	10.1	V	7
C18		6		4				
C19		25		27				
C20		7		9				
C21		36	32	34				
C22		16		14				
C23		37		39				
C24		18		20				
C25		23	13	22				
C26		5		4				
C27		26		27				
C28		8		9				
C29		35	33	34				
C30		15		14				
C31		38		39				
C32		19		20				

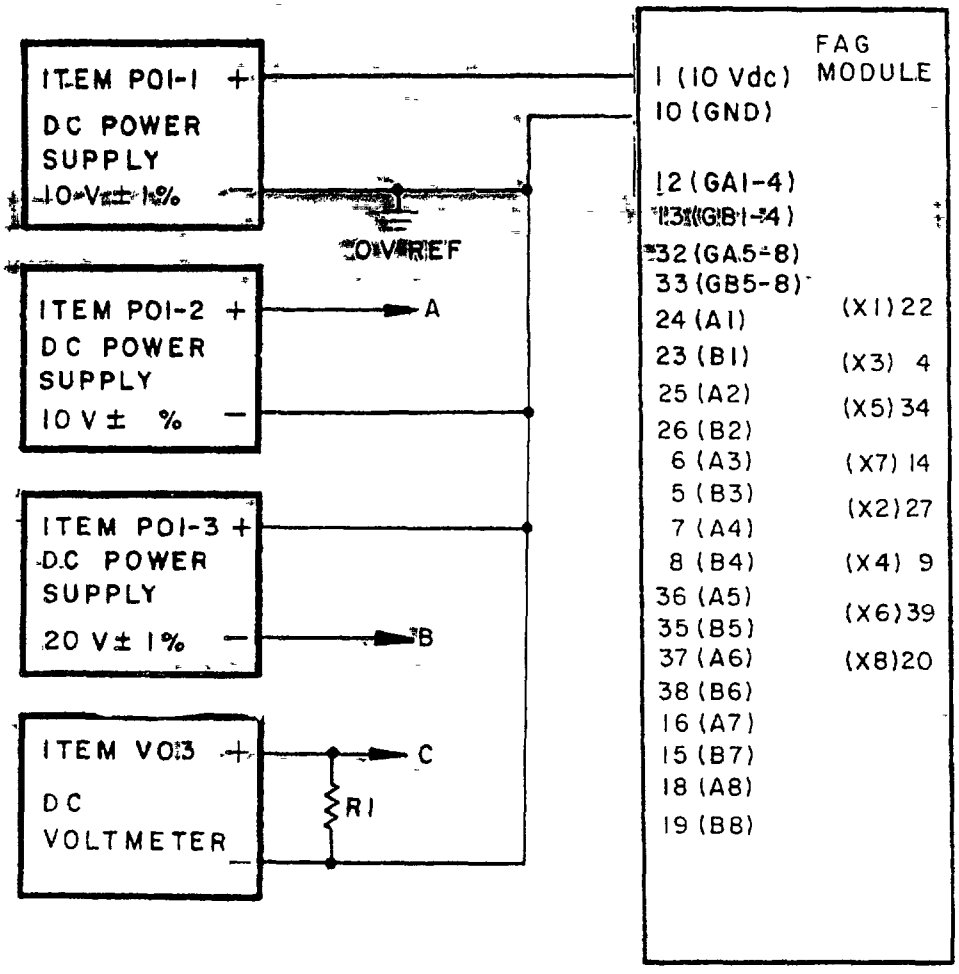
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:05Z
Check the source to verify that this is the current version before use.



NOTE. R1 = 10 kΩ ±1%, 1/8 W.

FIGURE 6. ON resistance check test circuit for test conditions C1-C16, table IV.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:05Z
Check the source to verify that this is the current version before use.



NOTE R1 = 10 kΩ ±1%, 1/8 W.

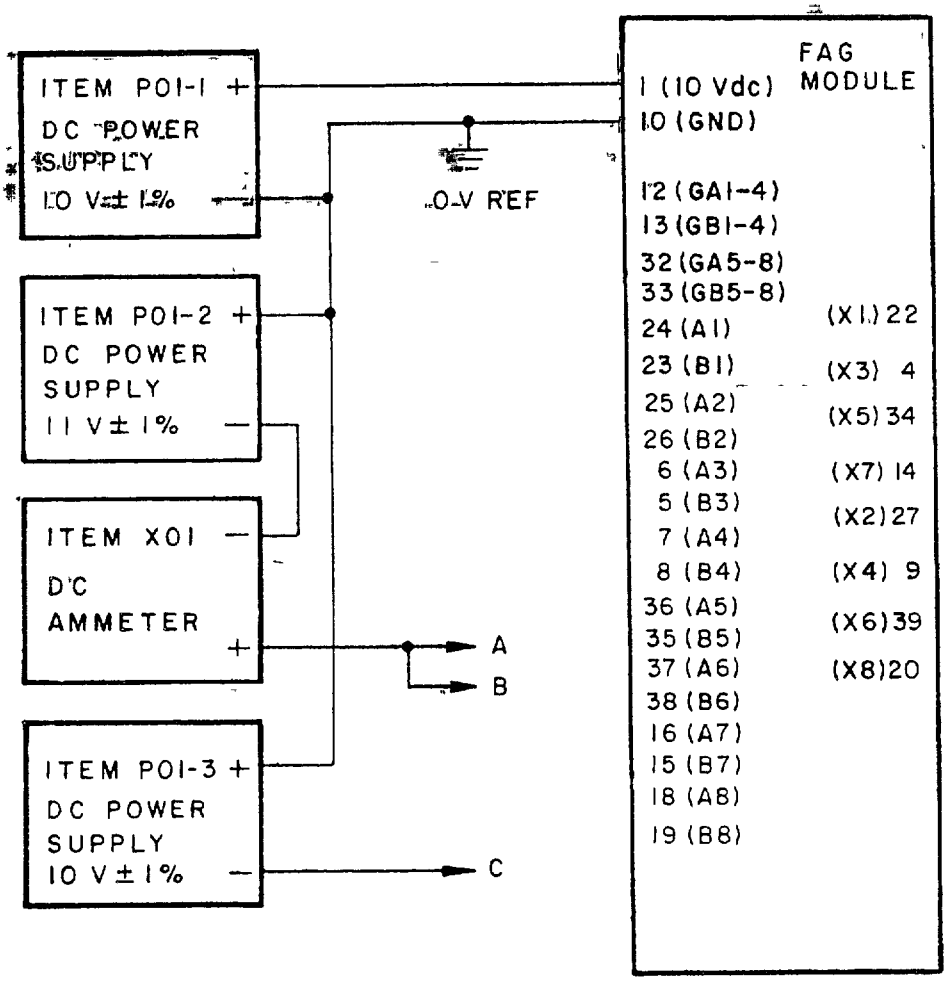
FIGURE 7. ON resistance check test circuit for test conditions C17-C32, table IV

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:05Z
Check the source to verify that this is the current version before use.

TABLE V. OFF resistance check test.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
D1	OFF resistance check	12	22	24	0.0	1.0	µA	8
D2	↓	↓	4	6	↓	↓	↓	↓
D3			27	25				
D4		↓	9	7				
D5		13	22	23				
D6		↓	4	5				
D7			27	26				
D8		↓	9	8				
D9		32	34	36				
D10		↓	14	16				
D11			39	37				
D12		↓	20	18				
D13		33	34	35				
D14		↓	14	15				
D15		↓	39	38				
D16	↓	↓	20	19	↓	↓	↓	↓

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:05Z
Check the source to verify that this is the current version before use.



CAUTION.

Do not isolate the chassis of ammeter XO1 from earth ground.

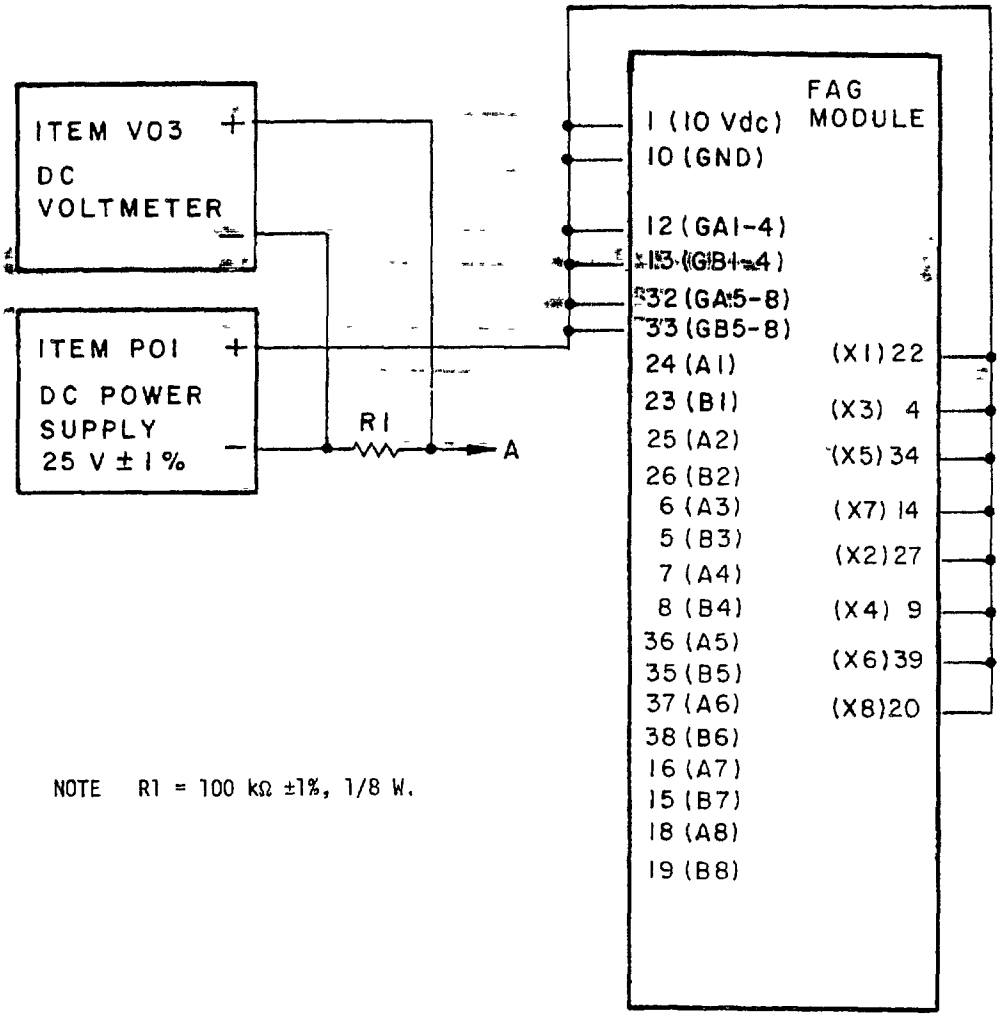
FIGURE 8 OFF resistance check test circuit for table V.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:05Z
Check the source to verify that this is the current version before use.

TABLE VI. Analog input leakage current check test.

Test condition	Parameter	Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
E1	Analog input leakage current check	24	0	50	mV	9
E2	↓	25	↓	↓	↓	↓
E3		6				
E4		7				
E5		36				
E6		37				
E7		16				
E8		18				
E9		23				
E10		26				
E11		5				
E12		8				
E13		35				
E14		38				
E15		15				
E16		19				

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:05Z
Check the source to verify that this is the current version before use.



NOTE R1 = 100 kΩ ±1%, 1/8 W.

FIGURE 9 Analog input leakage current check test circuit for table VI

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:05Z
Check the source to verify that this is the current version before use.

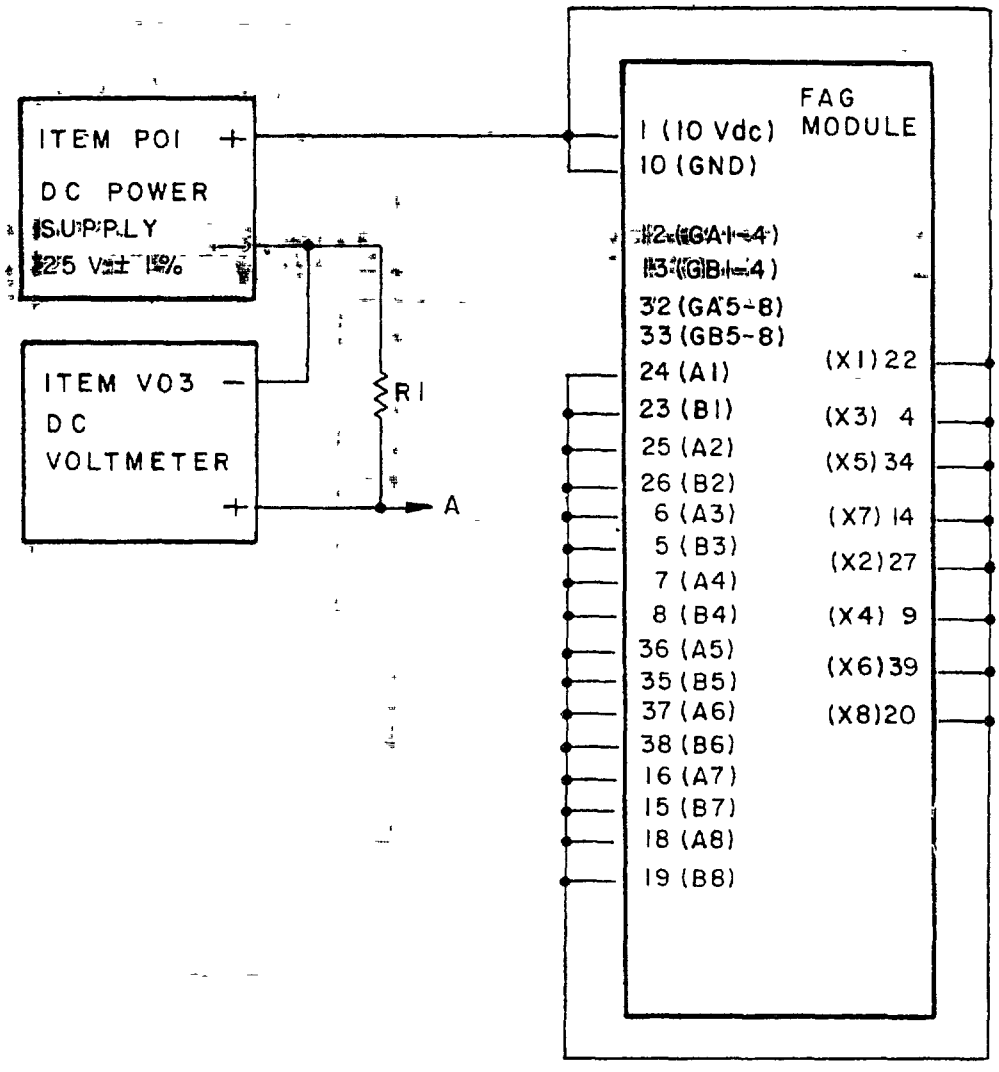
TABLE VII. Gate input leakage current check test.

Test condition	Parameter	Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
F1	Gate input leakage current check	12	0	10	mV	10
F2	↓	13	↓	↓	↓	↓
F3		32				
F4		33				

TABLE VIII Output leakage current check test.

Test condition	Parameter	Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
G1	Output leakage current check	22	0	50	mV	11
G2	↓	4	↓	↓	↓	↓
G3		34				
G4		14				
G5		27				
G6		9				
G7		39				
G8		20				

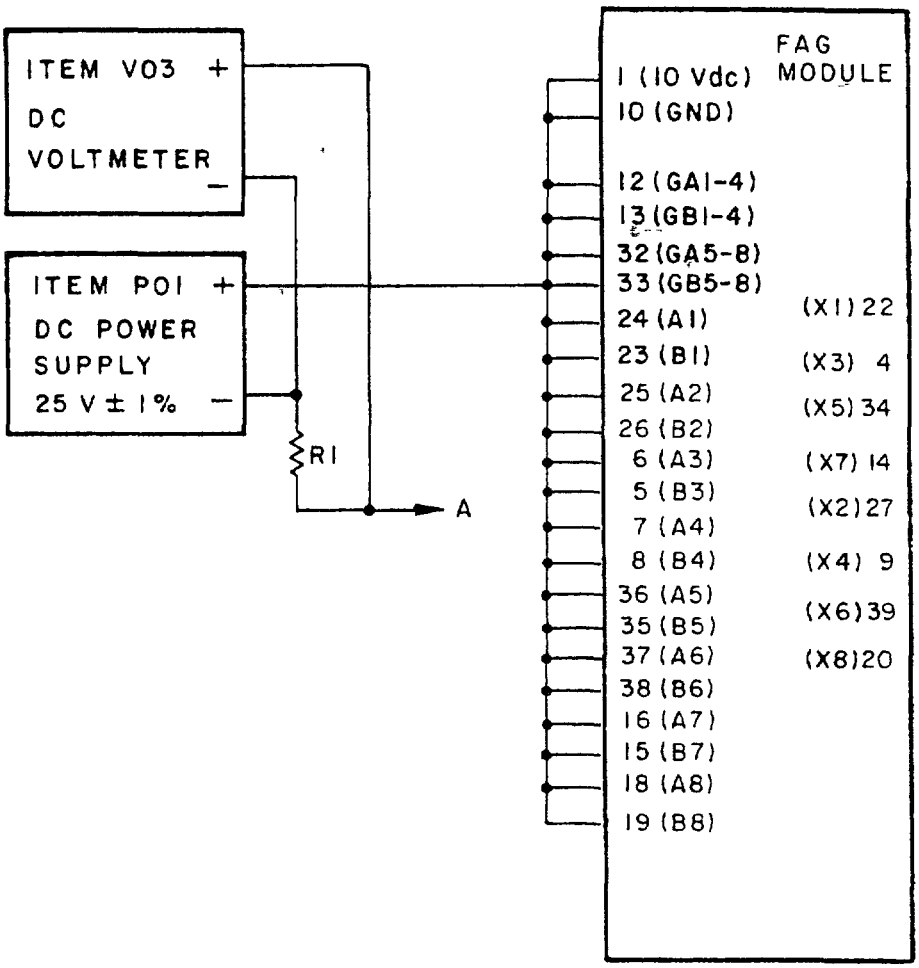
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:05Z
Check the source to verify that this is the current version before use.



NOTE. R1 = 100 kΩ ±1%, 1/8 W.

FIGURE 10 Gate input leakage current check test circuit for table VII.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:05Z
Check the source to verify that this is the current version before use.



NOTE. R1 = 100 kΩ ±1%, 1/8 W.

FIGURE 11 Output leakage current check test circuit for table VIII

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:05Z
Check the source to verify that this is the current version before use.

TABLE IX. Analog input, gate input, analog output, and gate to output capacitance test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
H1	Analog input capacitance	24	1,10	0	21	pF	12
H2		23					
H3		25					
H4		26					
H5		6					
H6		5					
H7		7					
H8		8					
H9		36					
H10		35					
H11		37					
H12		38					
H13		16					
H14		15					
H15		18					
H16		19					

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:05Z
Check the source to verify that this is the current version before use.

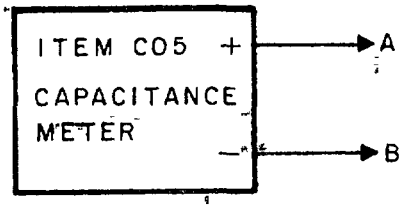
TABLE IX. Analog input, gate input, analog output, and gate to output capacitance test - Continued.

Test condition	Parameter	Connect A to pin	Connect B to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
J1	Gate input capacitance	12	1,10	0	30	pF	12
J2		13					
J3		32					
J4		33					
K1	Analog output capacitance	22			26		
K2		4					
K3		34					
K4		14					
K5		27					
K6		9					
K7		39					
K8		20					
L1	Gate to output capacitance	12	22		16		
L2		12	4				
L3		13	27				
L4		13	9				
L5		32	34				
L6		32	14				
L7		33	39				
L8		33	20				

TABLE X. Power supply transient rise time test.

Test condition	Parameter	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
M	Power supply transient rise time	2.5	7.5	µs	13,14

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:05Z
Check the source to verify that this is the current version before use.

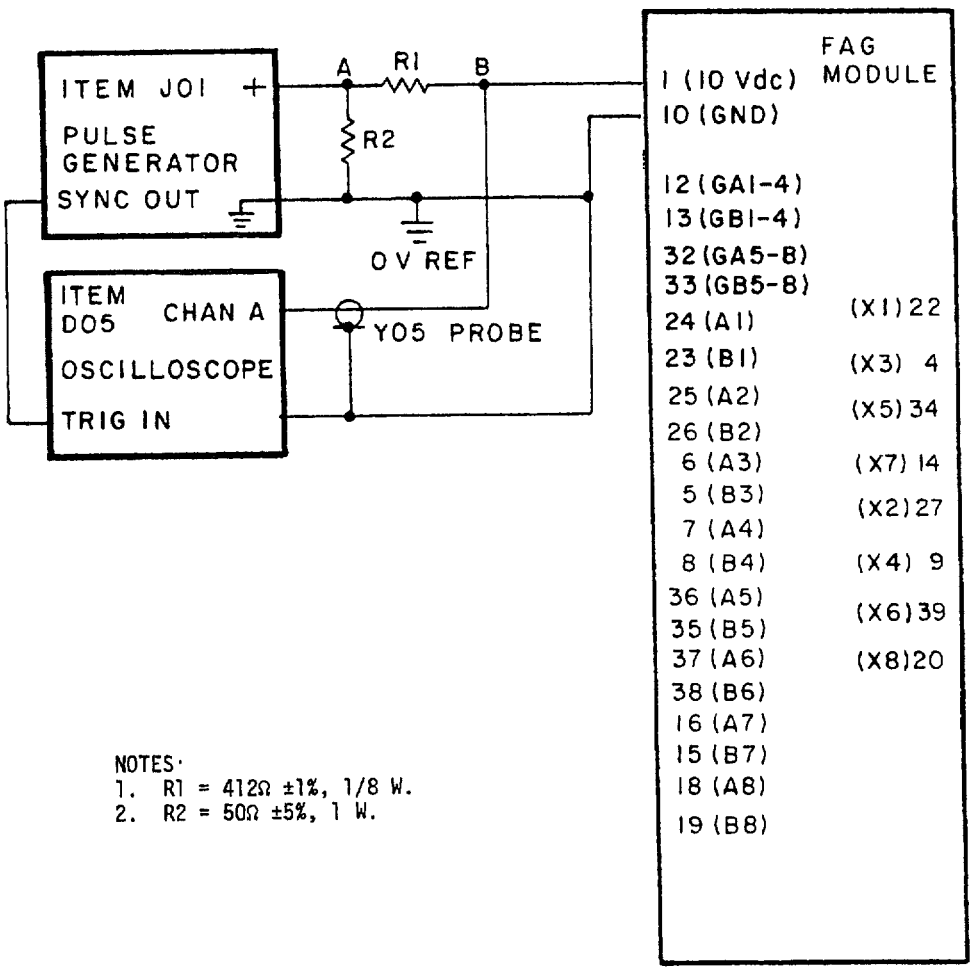


	FAG MODULE
1 (10 Vdc)	
10 (GND)	
12 (GA1-4)	
13 (GB1-4)	
32 (GA5-8)	
33 (GB5-8)	
24 (A1)	(X1) 22
23 (B1)	(X3) 4
25 (A2)	(X5) 34
26 (B2)	
6 (A3)	(X7) 14
5 (B3)	(X2) 27
7 (A4)	
8 (B4)	(X4) 9
36 (A5)	(X6) 39
35 (B5)	
37 (A6)	(X8) 20
38 (B6)	
16 (A7)	
15 (B7)	
18 (A8)	
19 (B8)	

NOTES:
1. Do not attach test connector or wires to any of the module pins other than pins under test Stray capacitance added to other pins may affect the reading.
2. Null out the effects of the test leads prior to connecting to the module

FIGURE 12. Analog input, gate input, analog output, and gate to output test circuit for table IX.

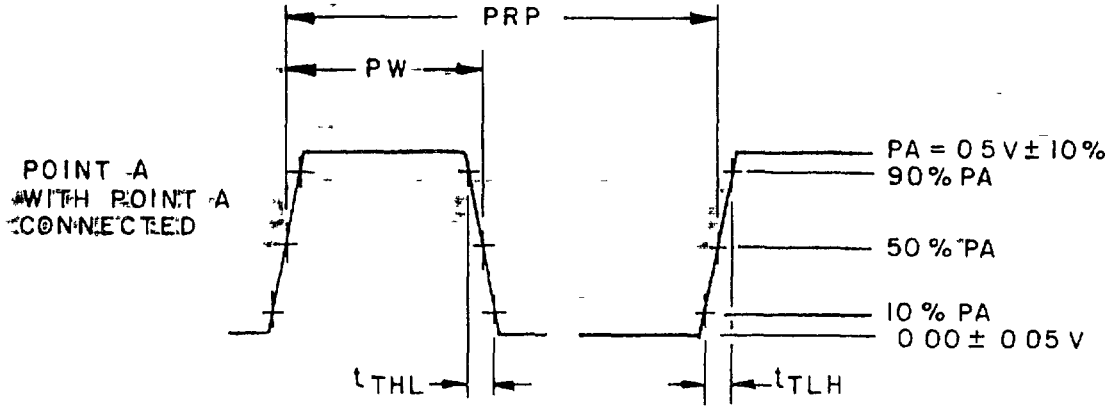
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:05Z
Check the source to verify that this is the current version before use.



NOTES:
1. R1 = 412Ω ±1%, 1/8 W.
2. R2 = 50Ω ±5%, 1 W.

FIGURE 13. Power supply transient rise time test circuit for table X

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:05Z
Check the source to verify that this is the current version before use.



NOTE Input pulse parameters
PW = 30 µs ±10%.
PRP = 60 µs ±10%.
$t_{TLH} = t_{THL}$ = 50 ns maximum

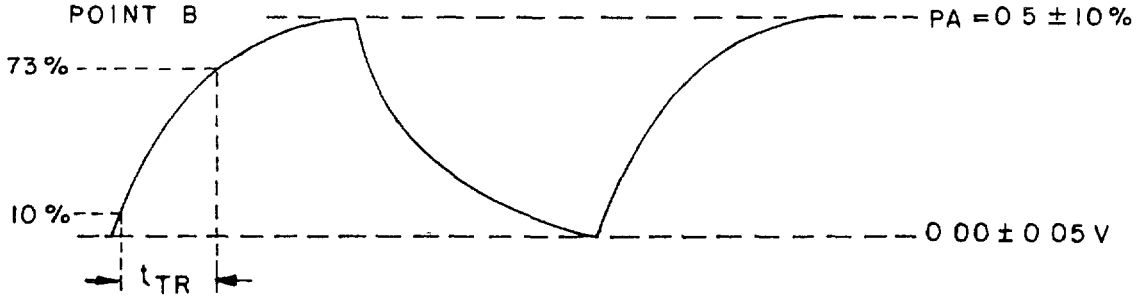


FIGURE 14. Power supply transient rise time test circuit waveforms for table X

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:05Z
Check the source to verify that this is the current version before use.

TABLE XI. Propagation delay time (t_{PLH} and t_{PHL}) test.

Test condition	Parameter	Connect					Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
N1	Propagation delay time (t_{PLH})	24	23	22	12	13	0	100	ns	15,16
N2	↓	6	5	4	12	13	↓	↓	↓	↓
N3		36	35	34	32	33				
N4		16	15	14	32	33				
N5		25	26	27	12	13				
N6		7	8	9	12	13				
N7		37	38	39	32	33				
N8		18	19	20	32	33				
P1	Propagation delay time (t_{PHL})	24	23	22	12	13				
P2	↓	6	5	4	12	13				
P3		36	35	34	32	33				
P4		16	15	14	32	33				
P5		25	26	27	12	13				
P6		7	8	9	12	13				
P7		37	38	39	32	33				
P8	↓	18	19	20	32	33	↓	↓	↓	↓

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:05Z
Check the source to verify that this is the current version before use.

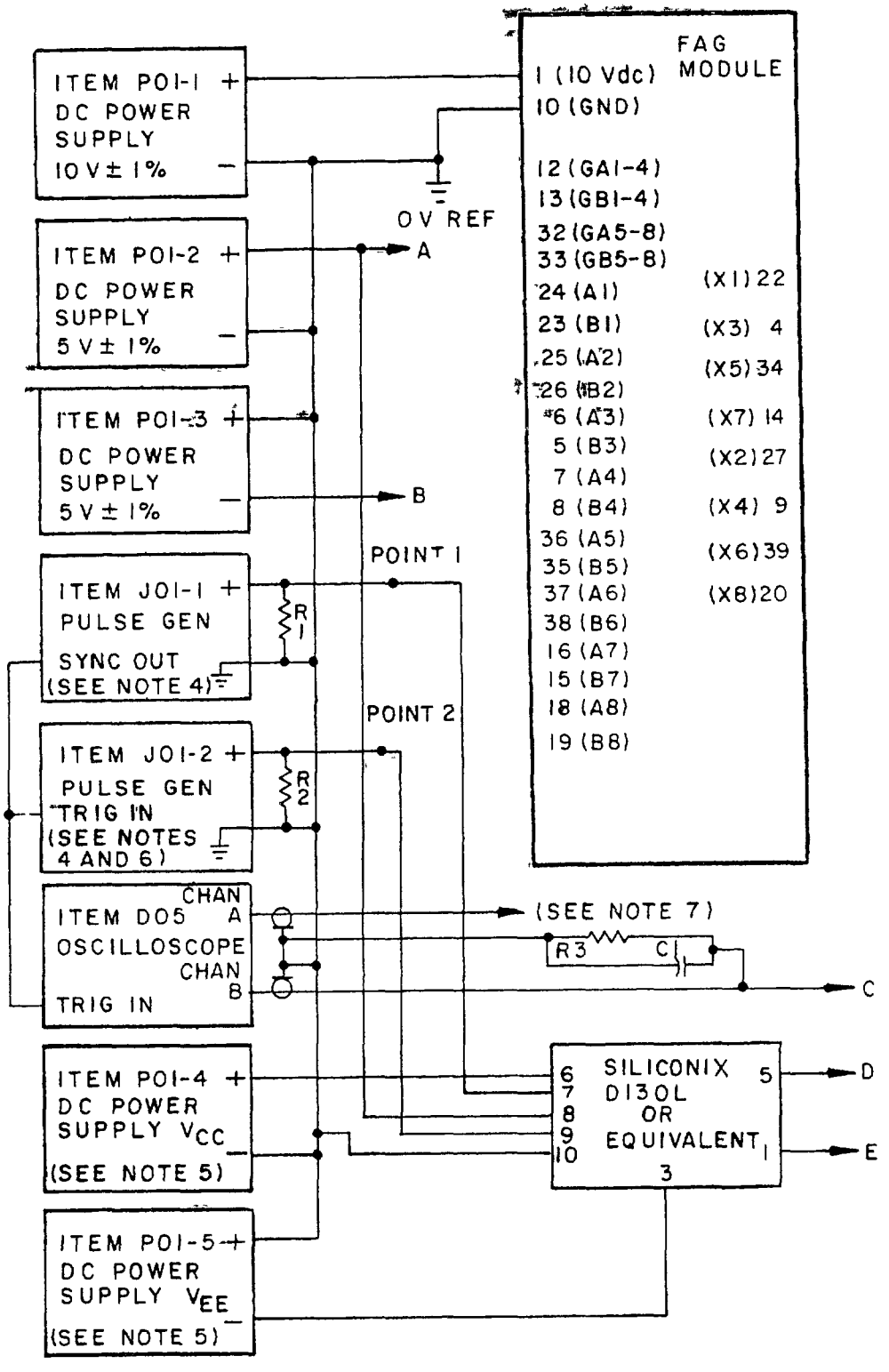
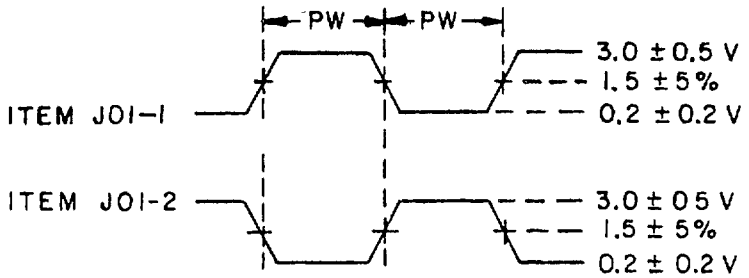


FIGURE 15 Propagation delay time (t_{PLH} and t_{PHL}) test circuit for table XI.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:05Z
Check the source to verify that this is the current version before use.

NOTES:
1. R1 = R2 = 50Ω ±5%, 1 W.
2. R3 = 10 kΩ ±1%, 1/8 W.
3. C1 = 30 pF ±5% (including probe and parasitic capacitance).
4. Initially adjust items J01-1 and J01-2 as follows:

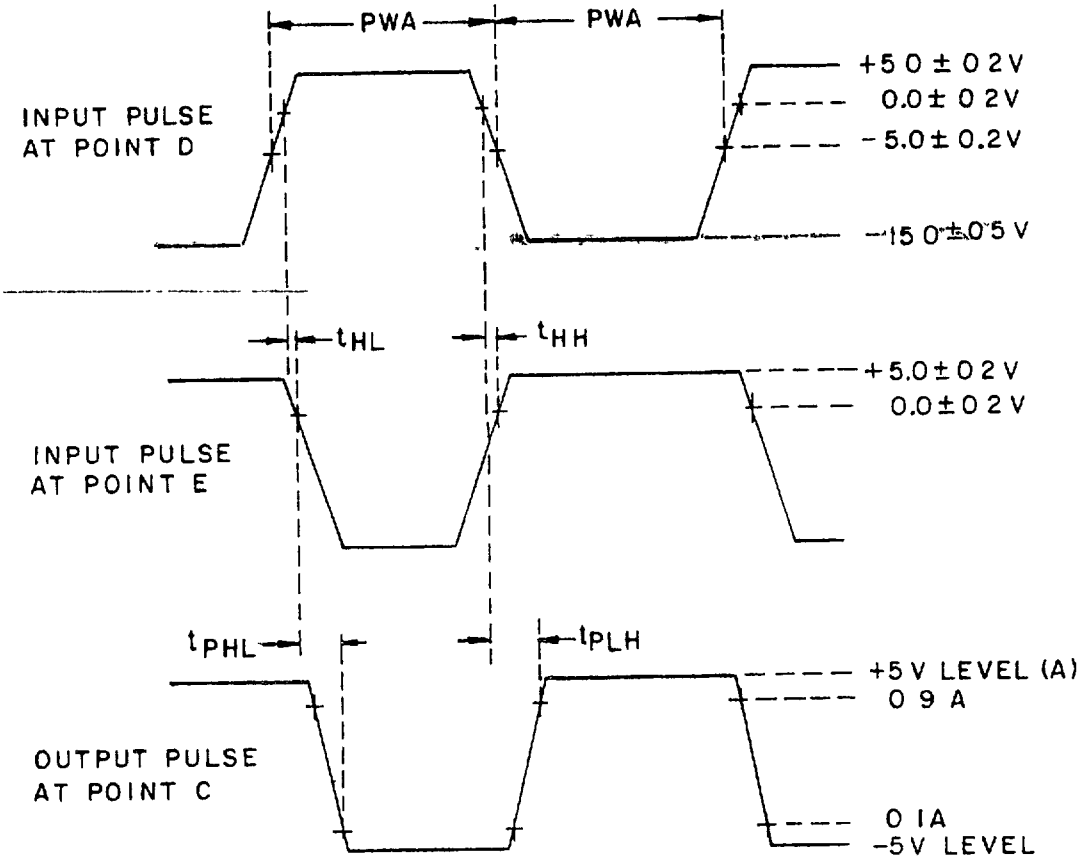


Input pulse parameters:
PW = 500 ±50 ns.
$t_{TLH} = t_{THL} = 25 \pm 10$ ns.

5. Initially adjust item P01-4 (V_{CC}) to 7 V and item P01-5 (V_{EE}) to -17 V. Adjust item P01-4 to obtain a +5.0 ±0.2 V and item P01-5 to obtain a -15.0 ±0.5 V level at points D and E as specified on figure 16.
6. Adjust item J01-2 pulse width and delay to obtain the specified delay between waveforms D and E ($t_{HL} = t_{HH} = 0 \pm 1$ ns) as shown on figure 16.
7. Connect item D05 channel A to point D for t_{PLH} measurements and to point E for t_{PHL} measurements.

FIGURE 15. Propagation delay time (t_{PLH} and t_{PHL}) test circuit for table XI - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:05Z
Check the source to verify that this is the current version before use.



NOTE: Input pulse parameters
PWA = 500 ±50 ns
$t_{TLH} = t_{THL}$ = 100 ±50 ns.
t_{HL} (hold time low) = t_{HH} (hold time high) = 0 ±1 ns.

FIGURE 16 Propagation delay time (t_{PLH} and t_{PHL}) test circuit waveforms for table XI.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:05Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS: This form is provided to solicit beneficial comments which may improve this document and enhance its use. DoD contractors, government activities, manufacturers, vendors, or other prospective users of the document are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity. A response will be provided to the submitter, when name and address is provided, within 30 days indicating that the 1426 was received and when any appropriate action on it will be completed.

NOTE This form shall not be used to submit requests for waivers, deviations or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

DOCUMENT IDENTIFIER *(Number)* AND TITLE
MIL-M-28787/113A (EC) SEM, Switch, Analog, Key Code FAG

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C REASON FOR RECOMMENDED CHANGE(S)

2. REMARKS

SUBMITTED BY *(Printed or typed name and address — Optional)*

TELEPHONE NO.

DATE

DD FORM 1426 1 OCT 76 EDITION OF 1 JAN 72 MAY BE USED UNTIL EXHAUSTED

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:05Z
Check the source to verify that this is the current version before use.

FOLD

DEFENSE LOGISTICS AGENCY

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

NO POSTAGE
NECESSARY
IF MAILED
IN THE
UNITED STATES

BUSINESS REPLY MAIL

FIRST CLASS PERMIT NO 4966 WASHINGTON D C

POSTAGE WILL BE PAID BY THE DEFENSE LOGISTICS AGENCY

Commander
Naval Electronic Systems Command
ATTN: ELEX 8111
Washington, DC 20360

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:05Z
Check the source to verify that this is the current version before use.